I can help you prepare a detailed SCI-style research manuscript on the solution combustion synthesis of copper nanorods or copper nanowires using copper nitrate and hexamethylenetetramine (HMTA), including the reaction chemistry, proposed experimental design, characterization methods, data tables, discussion, and references.

There is one important scientific distinction: example data must not be presented as real experimental results. I can provide clearly labeled hypothetical data for a manuscript-development exercise, but those data must be replaced with measured results before submission to an SCI-indexed journal.

There is also a chemistry issue worth addressing at the outset: copper nitrate is an oxidizer, while HMTA can act as a fuel and nitrogen-containing complexing agent. Calling HMTA a reducing agent in this system requires evidence; its role depends on the reaction conditions. Moreover, solution combustion does not automatically produce copper nanowires. The formation of metallic copper, copper oxides, nanoparticles, rods, or wires must be established experimentally.

1. What the existing literature suggests

The literature points to a promising research question, but also shows why the manuscript needs a clear novelty claim.

- A study of solution combustion using copper nitrate and HMTA investigated the fuel-to-oxidizer ratio and reported CuO nanoparticles, not a demonstrated nanowire product.

  ResearchGate

- An earlier Materials Letters study showed that changing the fuel content in copper-nitrate combustion can shift the products between copper oxides and metallic copper. Its fuel was carbohydrazide, not HMTA.

  ScienceDirect

- Copper nitrate and HMTA are also used in aqueous growth of copper oxide nanostructures, but that is a different process from solution combustion.

  The Journal of Physical Chemistry C

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Potential research gap: investigate whether combustion-derived copper-containing nuclei can be directed toward one-dimensional growth, and determine which phase and morphology actually form. A credible paper should not assume that nanowires will form simply because the precursors are suitable.

2. Proposed manuscript

The following is a manuscript-development draft. The numerical results later in the draft are explicitly illustrative placeholders—not measurements and not suitable for submission as experimental findings.

Solution Combustion Synthesis of Copper-Based One-Dimensional Nanostructures Using Copper Nitrate and Hexamethylenetetramine: Phase Formation, Morphology Control, and Growth Mechanism

Manuscript type: Original research article — experimental study proposal
Status: Draft for experimental validation; illustrative data are not experimental findings.

Abstract

One-dimensional copper-based nanostructures, including copper nanorods, copper nanowires, and copper oxide nanowires, are of interest for conductive networks, sensors, catalysis, and functional coatings. Solution combustion synthesis offers a potentially rapid route to copper-containing nanomaterials through the interaction of an oxidizing precursor and an organic fuel. In this work, copper(II) nitrate trihydrate is proposed as the copper precursor and oxidizer, while hexamethylenetetramine (HMTA) is used as the organic fuel. The study is designed to investigate how precursor composition, solution chemistry, thermal history, and post-synthesis treatment affect phase composition, particle morphology, and the formation of one-dimensional structures.

A systematic characterization strategy combining X-ray diffraction (XRD), scanning electron microscopy (SEM), transmission electron microscopy (TEM), energy-dispersive X-ray spectroscopy (EDS), and X-ray photoelectron spectroscopy (XPS) is proposed to distinguish metallic copper from Cu_2O and CuO. Morphological measurements will establish the proportion of rod-like and wire-like products and quantify their length, diameter, and aspect ratio. A central

objective is to determine whether the process produces genuine one-dimensional structures directly or whether subsequent oxidation, reduction, or recrystallization is required.

The study will provide a basis for evaluating the feasibility of HMTA-assisted solution combustion for copper-based nanostructures. Any claim of successful nanowire formation, enhanced yield, or improved performance will depend on experimental confirmation.

Keywords: solution combustion synthesis; copper nanowires; copper nanorods; copper nitrate; hexamethylenetetramine; copper oxide; morphology control; nanomaterials.

1. Introduction

One-dimensional nanomaterials possess a high aspect ratio and anisotropic transport properties that make them attractive for conductive coatings, flexible electronics, chemical sensors, and catalytic applications. Copper is especially interesting because of its relatively high electrical conductivity and lower material cost compared with silver. However, copper nanostructures can oxidize readily, and their morphology and phase composition are sensitive to synthesis conditions.

Copper nanowires have been prepared through several approaches, including solution-phase reduction, template-assisted growth, electrochemical deposition, and polyol-based synthesis. These methods can produce high-aspect-ratio structures, but may require specialized templates, surfactants, reducing agents, or extended growth periods.

Solution combustion synthesis is an alternative materials-processing route based on an exothermic reaction between an oxidizing precursor and a fuel. Copper nitrate can provide Cu^{2+} ions and nitrate-derived oxidizing species, while HMTA can serve as an organic fuel. Depending on the reaction environment and fuel-to-oxidizer balance, the resulting solid may contain CuO, Cu_2O, metallic copper, or mixtures of these phases.

Importantly, HMTA also participates in aqueous coordination and precipitation chemistry. Its hydrolysis can generate ammonia and formaldehyde, changing the local pH and influencing copper hydroxide or oxide formation. Consequently, a copper nitrate–HMTA system should not be interpreted as a simple, single-step reduction reaction without supporting evidence.

The principal challenge is to determine whether the rapid thermal event characteristic of combustion can produce or preserve anisotropic copper-based structures. Combustion-derived powders often contain aggregated particles rather than isolated nanowires. A convincing demonstration therefore requires both phase identification and direct microscopic evidence of one-dimensional morphology.

1.1 Research objectives

The proposed work has four objectives:

1. Determine the crystalline phases formed from copper nitrate and HMTA under controlled solution-combustion conditions.
2. Establish the relationship between precursor composition and the morphology of the recovered material.
3. Investigate whether rod-like or wire-like structures originate during combustion or during subsequent processing.
4. Develop a reproducible route toward copper-based one-dimensional nanostructures, if supported by experimental observations.

1.2 Research hypotheses

H1: Precursor composition and thermal history influence the relative amounts of Cu, Cu_2O, and CuO.

H2: Solution chemistry and precursor decomposition influence particle aggregation and the probability of anisotropic growth.

H3: The presence of rod-like particles does not necessarily imply that a continuous metallic copper nanowire network has formed.

These hypotheses will be evaluated using independent synthesis replicates and complementary characterization methods.

2. Materials and methods

2.1 Chemicals

Copper(II) nitrate trihydrate, $Cu(NO_3)_2 \cdot 3H_2O$, and hexamethylenetetramine, $C_6H_{12}N_4$, are the proposed starting materials. Deionized water is the solvent.

Reagent purity, supplier, batch number, and storage conditions should be recorded. If additional reagents are introduced during pH adjustment, washing, or post-treatment, their identities and concentrations must also be reported.

2.2 Experimental design

The experimental program should separate three variables that can otherwise become confounded:

- **Combustion chemistry:** precursor composition and the amount of organic fuel.
- **Solution chemistry:** pH, precursor concentration, and the time between mixing and heating.
- **Thermal history:** heating profile, reaction duration, atmosphere, and cooling conditions.

A staged experimental design is preferable to changing all variables simultaneously. Initial experiments should establish the product phases and combustion behavior before optimization for morphology begins.

The fuel-to-oxidizer ratio should be calculated from an explicit redox-equivalence method and reported together with the actual precursor quantities. A ratio defined only by mass should not be confused with a stoichiometric redox ratio.

Because nitrate–organic fuel mixtures can undergo rapid, strongly exothermic reactions and generate hot gases, the reaction scale, vessel geometry, ventilation, and thermal containment must be assessed by a qualified laboratory before experimentation. Do not heat or ignite a sealed mixture. Scale-up should not be attempted from small-scale observations alone.

2.3 Synthesis procedure and sample tracking

For each independent experiment, document the following:

1. The identity and measured amount of each precursor.
2. The solvent volume and final precursor concentration.
3. The order of addition, mixing method, and any observed precipitation.
4. The measured pH before heating, where applicable.
5. The actual heating and reaction history.
6. The appearance and mass of the recovered solid.
7. The washing, drying, storage, and any post-treatment procedures.

Each sample should have a unique identifier linked to its complete experimental record. At least three independent synthesis replicates should be used for the principal conditions intended for quantitative comparison.

The precise operating conditions must be established experimentally rather than inferred from the desired final morphology.

2.4 Control experiments

The following controls will help distinguish combustion effects from ordinary solution chemistry:

- **Copper precursor control:** evaluate the copper-containing precursor without the organic fuel, using a separately justified and safe thermal protocol.
- **HMTA solution control:** document the behavior of HMTA under the relevant solution conditions without copper nitrate.
- **Non-combustion solution control:** investigate whether copper-containing precipitates or anisotropic structures form under aqueous conditions without a combustion event.
- **Post-treatment control:** compare the as-produced material with any subsequently reduced or oxidized material.

A hydrothermal or precipitation control is not a direct substitute for combustion synthesis; it is included to test whether the observed morphology may have formed through solution growth rather than combustion.

3. Characterization methods

3.1 X-ray diffraction

XRD should be used to identify the crystalline phases and determine whether the sample is predominantly metallic copper, cuprous oxide, cupric oxide, or a multiphase mixture.

The analysis should report the radiation source, scan range, step size, instrumental settings, reference patterns, and phase-identification procedure. Rietveld refinement may be used to estimate phase fractions when data quality and the selected structural models permit it.

Crystallite size estimated from peak broadening is not equivalent to the diameter of an entire nanowire. Instrumental broadening and strain should be considered.

3.2 Scanning electron microscopy

SEM should establish the overall morphology, aggregation state, and distribution of particle sizes. Images should be collected from multiple representative regions of each sample rather than selecting only visually favorable areas.

For a rod-like or wire-like product, report the number of objects measured, their length and diameter distributions, the fraction classified as one-dimensional structures, and the criteria used to classify each object.

3.3 Transmission electron microscopy

TEM and high-resolution TEM should be used to determine whether the apparent wires are continuous structures or chains of joined nanoparticles. Selected-area electron diffraction and lattice-fringe analysis can provide additional information about crystallinity and phase.

EDS mapping may identify copper and oxygen distributions, but EDS alone cannot reliably distinguish Cu^{0}, Cu^{+}, and Cu^{2+}.

3.4 X-ray photoelectron spectroscopy

XPS should be used to investigate surface oxidation states. Copper 2p spectra, including satellite features, should be interpreted together with the Cu LMM Auger region when distinguishing metallic copper and Cu(I) species.

Because copper nanostructures can oxidize during handling, the storage atmosphere and time between synthesis and characterization should be documented.

3.5 Image analysis

Morphology should be quantified using a documented image-analysis workflow. The following parameters are recommended:

$$
AR=\frac{L}{D}
$$

where (AR) is the aspect ratio, (L) is the measured length, and (D) is the measured diameter.

Report the median, interquartile range, and number of objects measured. Length and diameter should be obtained from calibrated images, and objects with ambiguous boundaries should be handled according to predefined criteria.

A nanorod or nanowire designation should be based on measured aspect ratio and morphology, not appearance alone. The terminology and classification threshold should be defined before analyzing the final dataset.

4. Reaction chemistry and proposed formation mechanism

The nitrate-containing precursor supplies copper ions and oxidizing species. HMTA is an organic fuel under combustion conditions, but its behavior in aqueous solution is more complex than that of a simple electron donor.

The overall reaction pathway may include:

1. Evaporation of water and concentration of the dissolved precursors.
2. Thermal decomposition of nitrate-containing species and HMTA.
3. Formation of a hot, reactive environment with gaseous products.
4. Nucleation of copper-containing crystalline phases.
5. Aggregation, coarsening, or possible anisotropic growth during the thermal event and cooling.
6. Surface oxidation or phase transformation during recovery and storage.

The relative contributions of these steps depend on the experimental conditions. The final phase cannot be predicted from the precursor names alone.

A possible pathway toward one-dimensional morphology would involve anisotropic crystal growth or directional assembly of smaller particles. This remains a hypothesis until microscopy and crystallographic evidence demonstrate the relevant structure. If the product is predominantly equiaxed CuO or Cu_2O, the manuscript should report that result rather than describe it as copper nanowire synthesis.

5. Illustrative data framework

The following values are hypothetical examples for designing tables and analysis. They are not measured results, literature-derived measurements, or evidence that this synthesis produces nanowires. They must not be reported as experimental findings.

Table 1. Example of an experimental-record table

Parameter	Sample A	Sample B	Sample C
Fuel-to-oxidizer ratio	To be measured/calculated	To be measured/calculated	To be measured/calculated
Initial pH	To be measured	To be measured	To be measured
Thermal history	Record actual profile	Record actual profile	Record actual profile
Recovered mass	Measure	Measure	Measure
Major crystalline phase	XRD result	XRD result	XRD result
Dominant morphology	SEM/TEM result	SEM/TEM result	SEM/TEM result

Table 2. Hypothetical morphology data for demonstrating calculations

Illustrative sample	Median length (nm)	Median diameter (nm)	Approximate aspect ratio
Example A	300	100	3
Example B	800	80	10
Example C	1,500	75	20

These values are mathematical examples only. They do not establish that the samples are copper, that they were produced by combustion, or that any of the structures were observed.

For real samples, the aspect ratio should be calculated for each measured object and the distribution reported. Dividing the median length by the median diameter provides a useful summary, but it is not necessarily equal to the median of the individual aspect ratios.

Table 3. Example phase-analysis reporting format

Sample	Cu phase fraction	Cu_2O phase fraction	CuO phase fraction	Method
Sample A	Measure	Measure	Measure	XRD refinement
Sample B	Measure	Measure	Measure	XRD refinement
Sample C	Measure	Measure	Measure	XRD refinement

Phase fractions must be determined from actual diffraction data using a stated analysis method. They must not be estimated from color, morphology, or precursor composition.

6. Results and discussion: analysis plan

6.1 Phase composition

The first question is whether the reaction produces metallic copper or copper oxide. If XRD identifies CuO as the dominant phase, the product should be described as copper oxide rather than metallic copper nanowires.

If multiple phases are detected, their relative proportions should be quantified where feasible. XPS and electron microscopy can help investigate surface oxidation and local phase differences, but their results should not be substituted directly for bulk phase fractions.

6.2 Morphology evolution

SEM and TEM results should be analyzed to distinguish isolated rods, continuous wires, elongated aggregates, and irregular particles. A convincing nanowire claim requires consistent evidence across multiple regions and samples.

The proposed mechanism should be supported by comparisons between the combustion product and the relevant controls. For example, if the non-combustion control produces similar rods, the evidence would suggest that solution chemistry, rather than the combustion event alone, is important to morphology formation.

6.3 Relationship between morphology and phase

A high aspect ratio does not establish metallic conductivity. Copper oxide wires and metallic copper wires can have different electrical behavior, so phase and morphology must be considered together.

If the objective is a conductive copper network, the study should also assess oxidation stability, electrical resistance, and the effect of contact resistance between individual structures.

6.4 Reproducibility and statistics

For the main synthesis conditions, report independent-batch variability rather than treating multiple objects from one micrograph as independent synthesis replicates.

Recommended reporting includes mean or median dimensions, dispersion, sample count, phase fractions, product yield, and uncertainty. Statistical comparisons should use methods appropriate to the distribution and experimental design.

7. Potential applications

If metallic copper nanowires are successfully obtained and their conductivity and stability are confirmed, potential applications include transparent conductive coatings, printed conductive networks, flexible electronics, and electromagnetic shielding.

Copper oxide nanostructures may instead be relevant to gas sensing, catalysis, photocatalysis, or electrochemical devices. The appropriate application should be selected according to the measured phase composition and functional properties.

An application claim should be supported by a direct test—for example, sheet resistance and optical transmission for a transparent conductor, or sensitivity and response time for a gas sensor.

8. Conclusions

This proposed study examines the feasibility of using copper nitrate and HMTA in solution combustion synthesis to produce copper-based one-dimensional nanostructures. The principal scientific challenge is to establish the relationship between precursor chemistry, reaction conditions, crystalline phase, and morphology.

The proposed characterization workflow combines XRD, SEM, TEM, EDS, and XPS to distinguish metallic copper from copper oxides and to verify whether observed elongated structures are genuine nanorods or nanowires. Control experiments are necessary to determine whether one-dimensional growth occurs during combustion or through a separate solution-mediated process.

No conclusion regarding nanowire yield, phase purity, optimal composition, or performance can be drawn until experimental measurements are available. If the reaction produces predominantly copper oxide nanoparticles, that result should guide a revised research question rather than be represented as successful metallic copper nanowire synthesis.

References

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Before submission, verify all author lists, journal details, page numbers, DOIs, and the exact claims against the original articles. The manuscript should also be checked against more recent work on copper nanowire growth and solution combustion.

3. What would make this a publishable SCI paper?

The most important contribution would not simply be using copper nitrate and HMTA together. That combination has already been investigated for copper oxide combustion synthesis.


ResearchGate

A stronger contribution would be a demonstrated and reproducible relationship between reaction conditions and one-dimensional morphology, supported by:

- Phase-pure or quantitatively characterized products.
- SEM and TEM evidence of genuine nanorods or nanowires.
- A justified formation mechanism supported by controls.
- Batch-to-batch reproducibility.
- A useful functional measurement, such as electrical conductivity or sensor performance.

One final clarification will materially change the experimental plan: Do you want the final product to be metallic copper (Cu) nanowires, or copper oxide (CuO) nanowires? These are different synthesis targets, and the manuscript should focus on one of them.